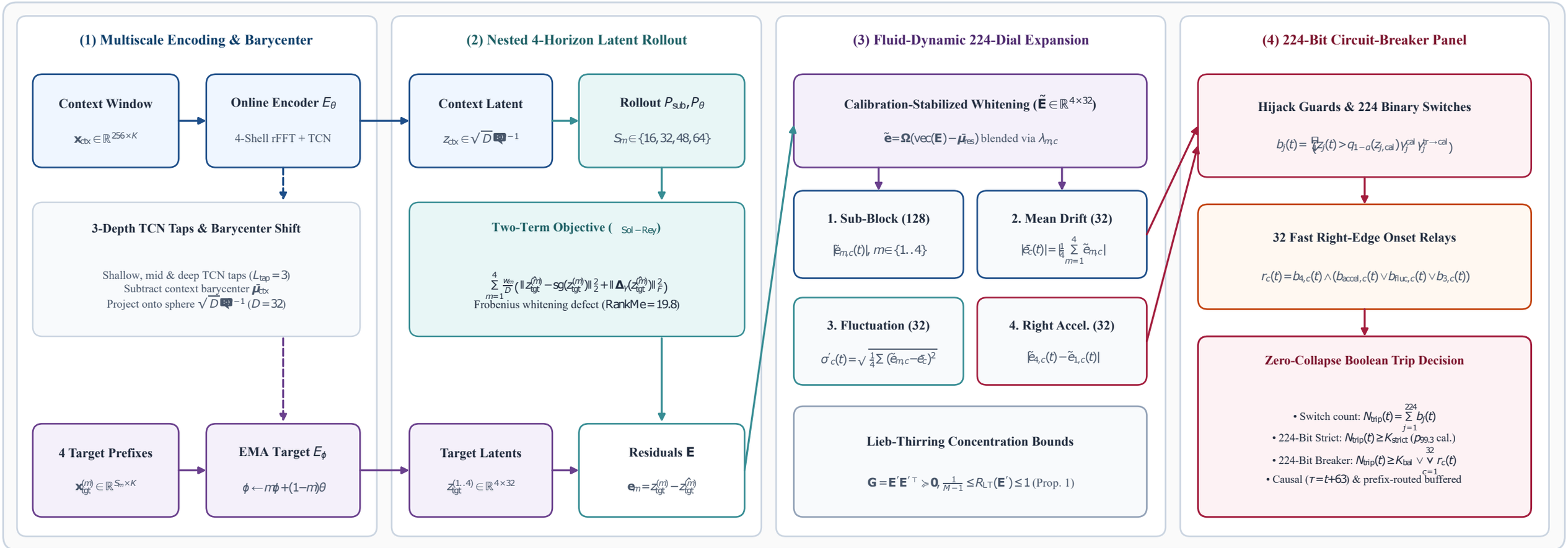

(1) Multiscale Encoding & Barycenter
Context Window
$\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{256\times K}$
Online Encoder E_θ
4-Shell rFFT + TCN
3-Depth TCN Taps & Barycenter Shift
Shallow, mid & deep TCN taps ($L_{\text{tap}}=3$)
Subtract context barycenter $\bar{\mu}_{\text{ctx}}$
Project onto sphere $\sqrt{D}\,\mathbb{S}^{D-1}$ ($D=32$)
4 Target Prefixes
$\mathbf{x}_{\text{tgt}}^{(m)} \in \mathbb{R}^{S_m\times K}$
EMA Target E_ϕ
$\phi \leftarrow m\phi+(1-m)\theta$
(2) Nested 4-Horizon Latent Rollout
Context Latent
$z_{\text{ctx}} \in \sqrt{D}\,\mathbb{S}^{D-1}$
Rollout P_{sub}, P_θ
$S_m \in \{16, 32, 48, 64\}$
Two-Term Objective (Sol – Rey)
$\sum_{m=1}^{4} \frac{w_m}{D}\left(\|\hat{z}_{\text{tgt}}^{(m)} - \text{sg}(z_{\text{tgt}}^{(m)})\|_2^2 + \|\Delta_\gamma(\hat{z}_{\text{tgt}}^{(m)})\|_F^2\right)$
Frobenius whitening defect (RankMe = 19.8)
Target Latents
$z_{\text{tgt}}^{(1..4)} \in \mathbb{R}^{4\times 32}$
Residuals $\mathbf{E}$
$\mathbf{e}_m = z_{\text{tgt}}^{(m)} - \hat{z}_{\text{tgt}}^{(m)}$
(3) Fluid-Dynamic 224-Dial Expansion
Calibration-Stabilized Whitening ($\tilde{\mathbf{E}} \in \mathbb{R}^{4\times 32}$)
$\tilde{\mathbf{e}} = \boldsymbol{\Omega}(\text{vec}(\mathbf{E}) - \bar{\boldsymbol{\mu}}_{\text{res}})$ blended via $\lambda_{m,c}$
1. Sub-Block (128)
$|\tilde{e}_{m,c}(t)|,\ m \in \{1..4\}$
2. Mean Drift (32)
$|\bar{e}_c(t)| = \left|\frac{1}{4}\sum_{m=1}^{4} \tilde{e}_{m,c}\right|$
3. Fluctuation (32)
$\sigma'_c(t) = \sqrt{\frac{1}{4}\sum(\tilde{e}_{m,c} - \bar{e}_c)^2}$
4. Right Accel. (32)
$|\tilde{e}_{4,c}(t) - \tilde{e}_{1,c}(t)|$
Lieb-Thirring Concentration Bounds
$\mathbf{G} = \mathbf{E}'\mathbf{E}'^{\top} \succcurlyeq \mathbf{0},\ \frac{1}{M-1} \le R_{\text{LT}}(\mathbf{E}') \le 1$ (Prop. 1)
(4) 224-Bit Circuit-Breaker Panel
Hijack Guards & 224 Binary Switches
$b_j(t) = \mathbb{1}\{z_j(t) > q_{1-\alpha}(z_{j,\text{cal}})\, v_j^{\text{cal}}\, v_j^{\text{tr}\to\text{cal}}\}$
32 Fast Right-Edge Onset Relays
$r_c(t) = b_{4,c}(t) \wedge (b_{\text{accel},c}(t) \vee b_{\text{fluc},c}(t) \vee b_{3,c}(t))$
Zero-Collapse Boolean Trip Decision
• Switch count: $N_{\text{trip}}(t) = \sum_{j=1}^{224} b_j(t)$
• 224-Bit Strict: $N_{\text{trip}}(t) \ge K_{\text{strict}}$ ($p_{99.3}$ cal.)
• 224-Bit Breaker: $N_{\text{trip}}(t) \ge K_{\text{bal}} \vee \bigvee_{c=1}^{32} r_c(t)$
• Causal ($\tau = t+63$) & prefix-routed buffered